Computer Engineering Department
Program: Sem VI
Course: Cloud Computing Lab(CSL605)
PART A
(PART A: TO BE COMPLETED BY STUDENTS)
Experiment No.5

A.1 Aim:

To demonstrate and implement Storage as a service using AWS S3 Service

A.2 Prerequisite:
Knowledge of Networking, Distributed Computing and knowledge of Software architectures.

A.3 Objective:
Objectives this experiment is to provide students overview of AWS storages, its Features and Services.

A.4 Outcome: (LO3)
After successful completion of this experiment student will be able to; Implement IAAS deployment model of cloud.

A.5 Theory:

Amazon Elastic Block storage (EBS): Storage in the form of blocks, each of these blocks associated with one particular instance, so when to access this block make sure that you have an instant connected to it and storage is accessed through that instant only.

Amazon Elastic File System (EFS): It is shared file system; hence it is not attached to a particular system or operating system.

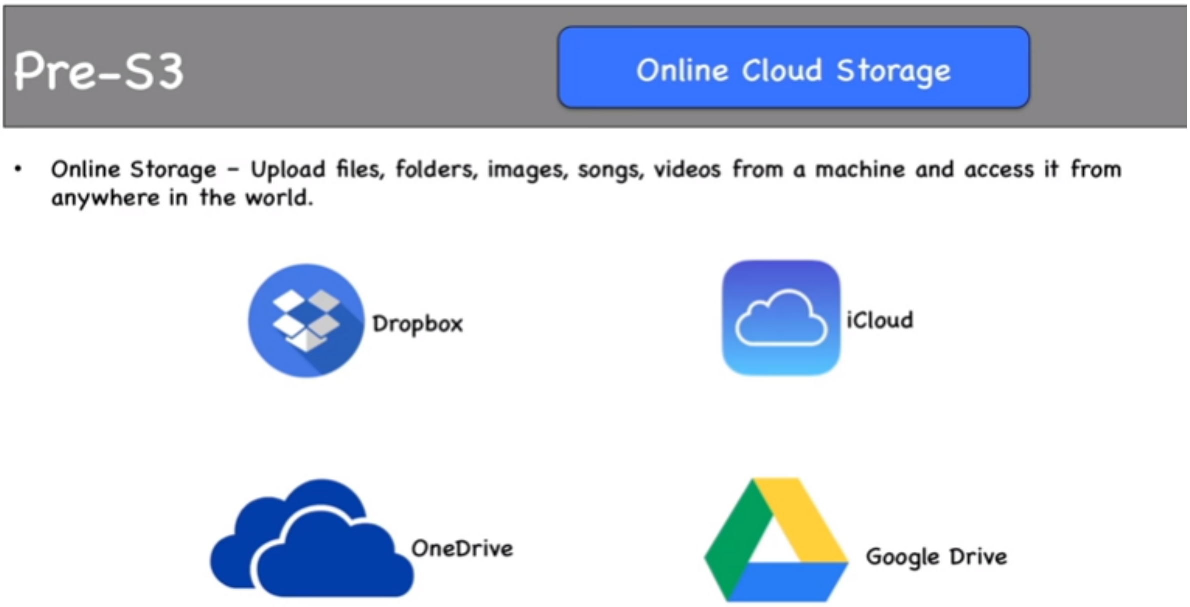
Amazon S3 Glacier: Need to store archive data, certain data that we would not want to retrieve or access on daily basis or frequently, such data you can put on **S3 Glacier.** That is it locks data for certain time during which you cannot access it, once you clear that duration you are free to access that data.
This storage is very cheap as compared to other data storage.
Example: Hospital System (birth certificate data): Birth certificate given by hospital once

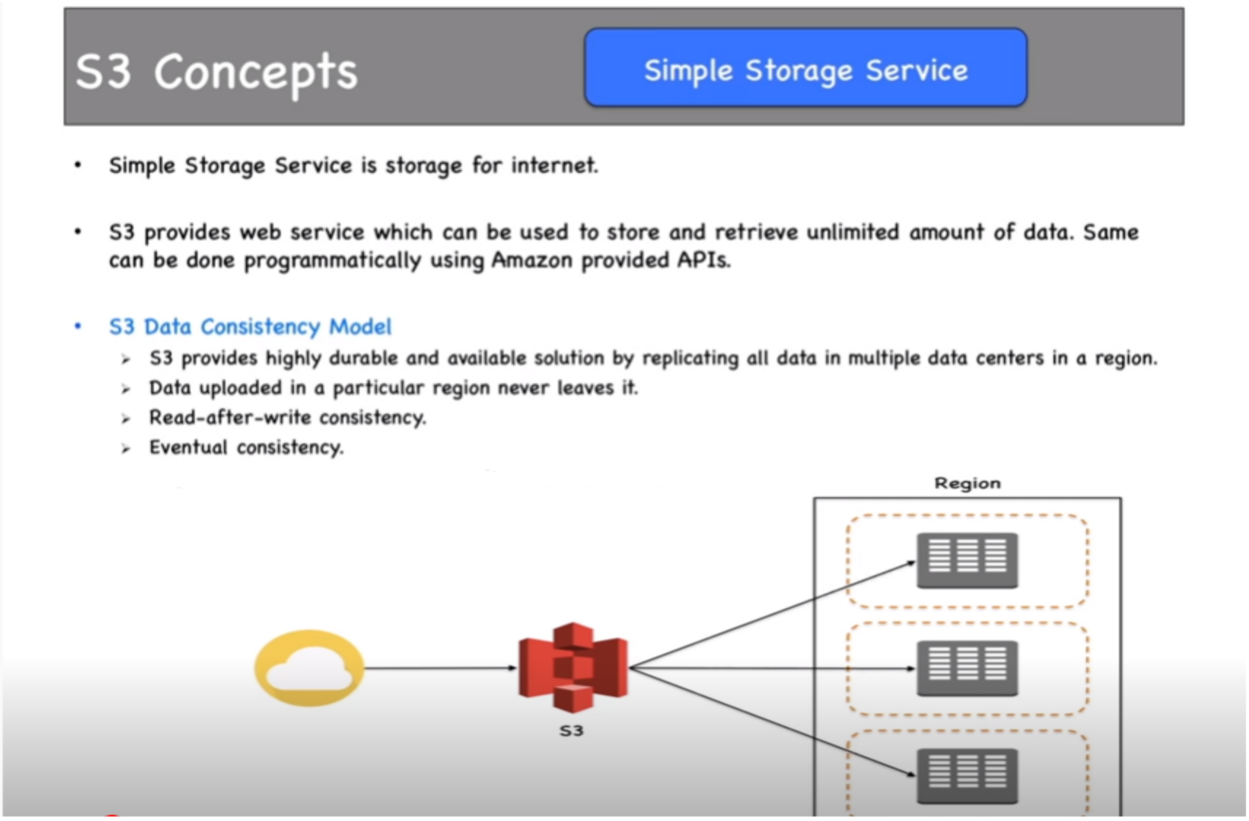
baby takes birth, so once you get it you are not requesting it again and again and hospital need to maintain it. That is data which is important but not needed on daily basis.

AWS Storage Gateway: Act as middle ware to move data from one system to another system.

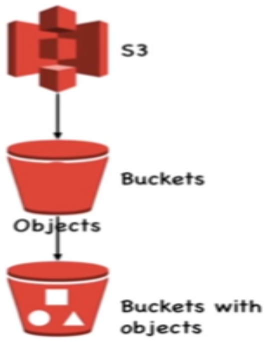
What is Amazon S3 (Simple Storage Service)

1) Amazon s3 has a simple web service interface that you can use to store and retrieve any amount of data, at anytime from anywhere on the web.
2) **S3 works on objects and buckets** (Bucket is an container and an object (doc, image, file etc) is an file, which you can stored in container)
3) Online cloud storage



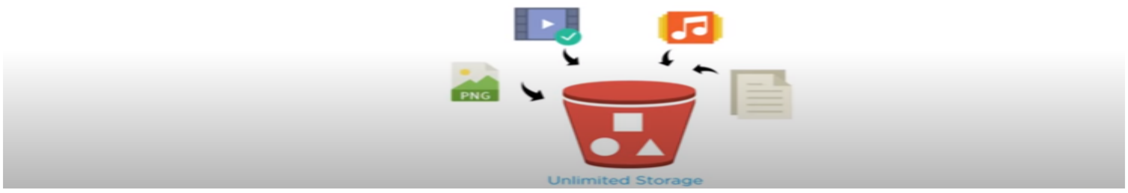


- S3 follows a storage hierarchy in keeping data (documents, images, videos, files etc.).



- Management console or S3 APIs can be used to manage buckets and objects.
- Bucket names have to be Globally unique irrespective of which region they are created in.
- Max 100 buckets can be created per account.

Amazon S3 (Simple Storage Service) provides object storage which is built for storing and recovering any amount of information or data from anywhere over the internet



4)

- ✓ Amazon S3 provides storage through web services interface
- ✓ It is designed for developers where web-scale computing can be easier for them
- ✓ It provides 99.999999999% durability and 99.99% availability of objects
- ✓ It can store computer files up to 5 terabytes in size

5)

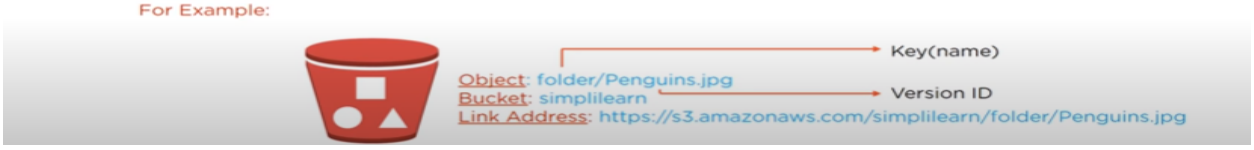
Object & Bucket in Amazon S3

An object consists of data, key(assigned name) and metadata

A bucket stores objects

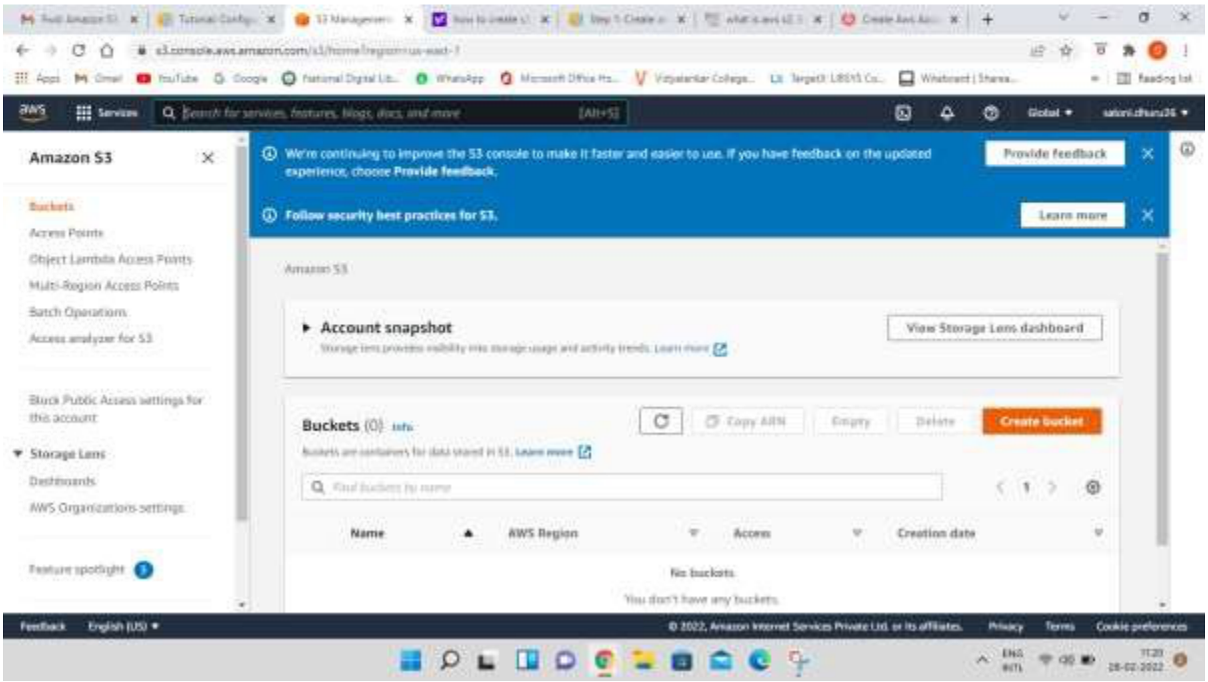
When data is added to the bucket, Amazon S3 creates a unique version ID and allocates it to the object

For Example:

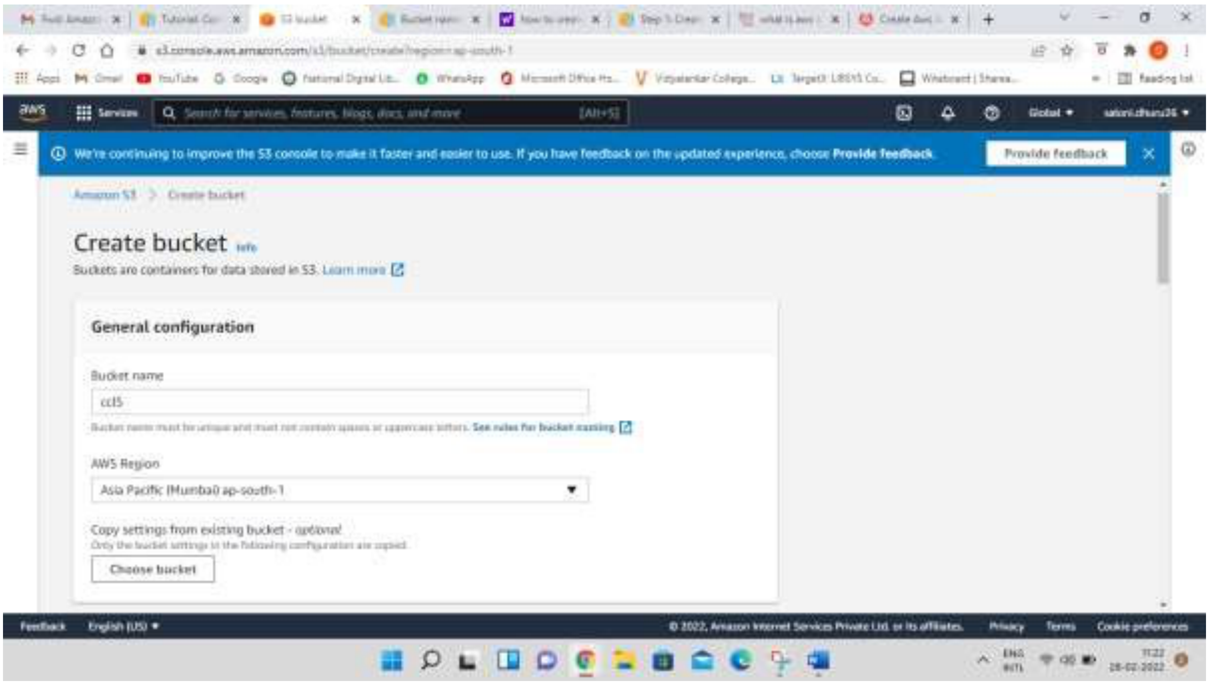


Steps to Implement Storage as a Service using Own Cloud/ AWS, Glaciers

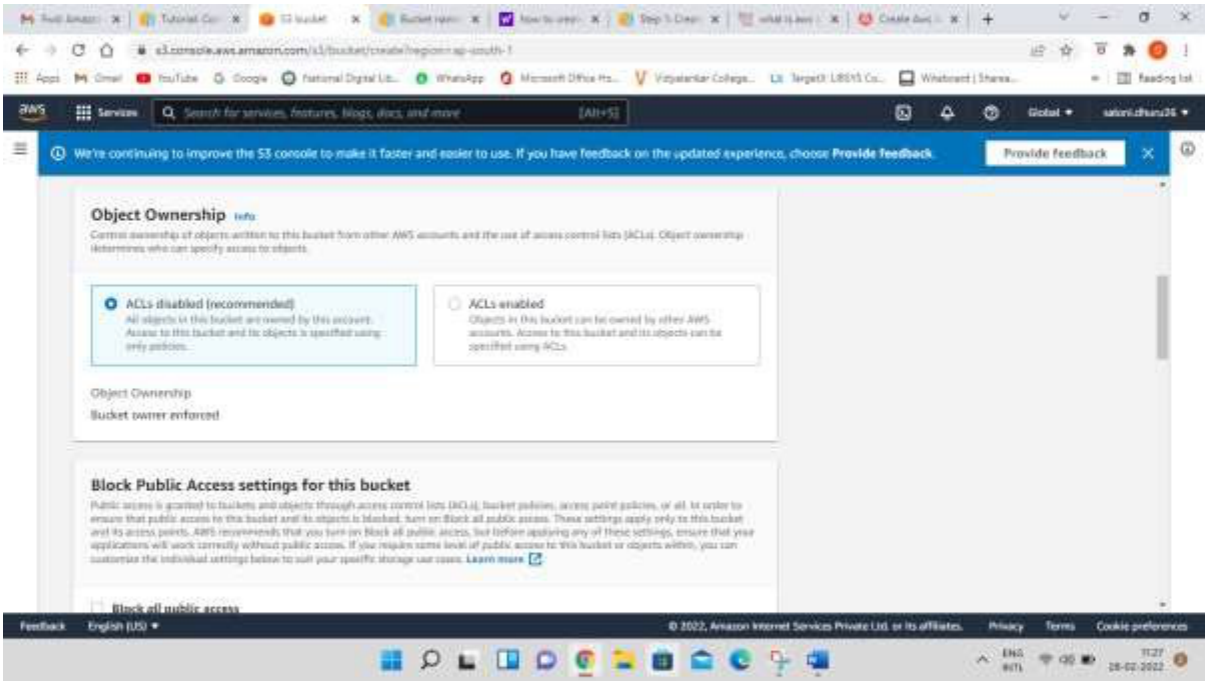
Step-1: click on create bucket



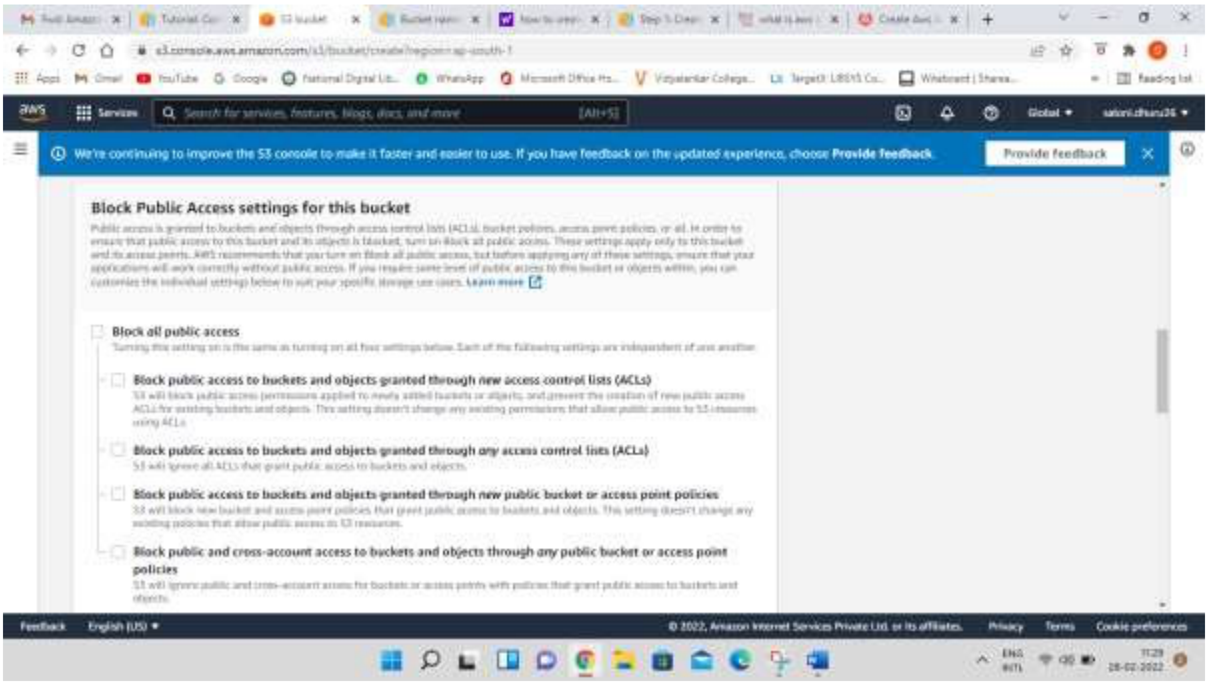
Step-2: Give Bucket name & select region for storage



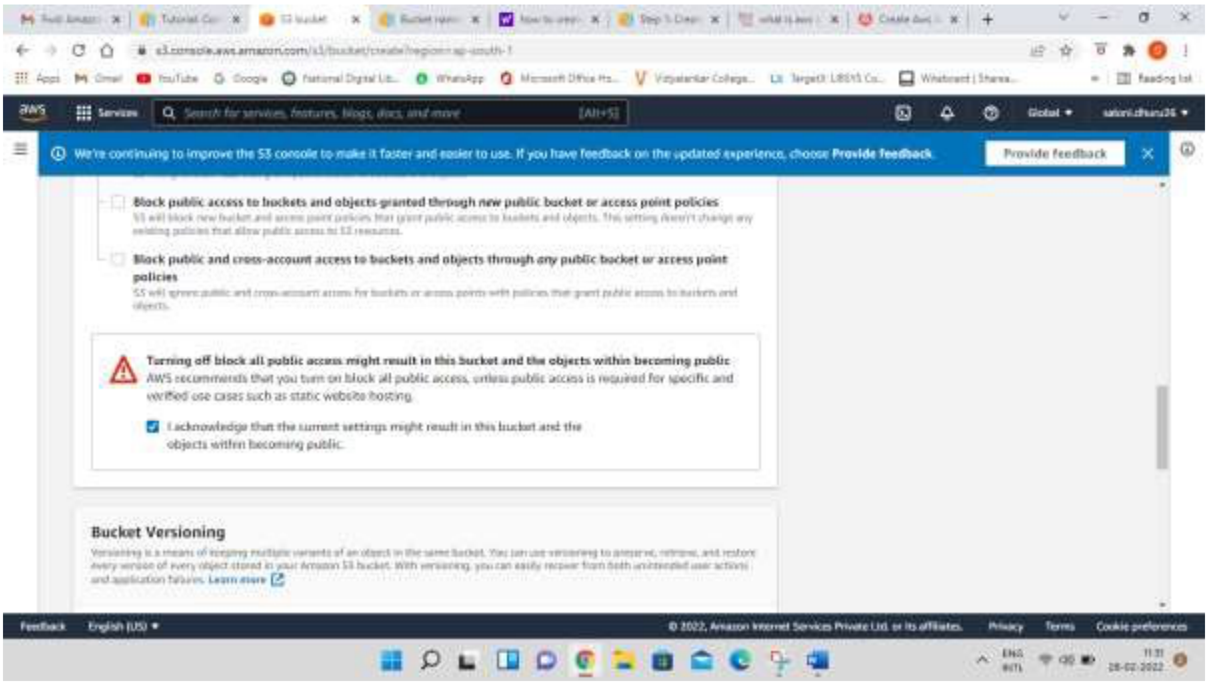
Step-3: Keep object ownership setting as ACLs Disabled as by-default



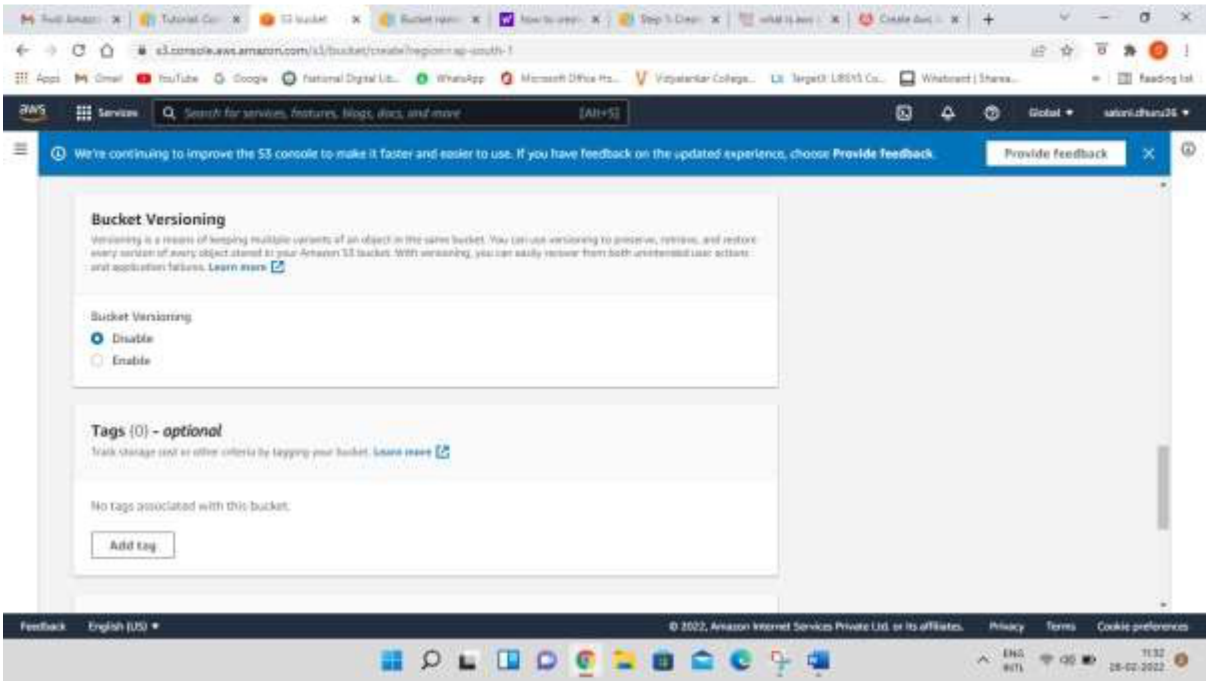
Step-4: Disable block all public access checkbox



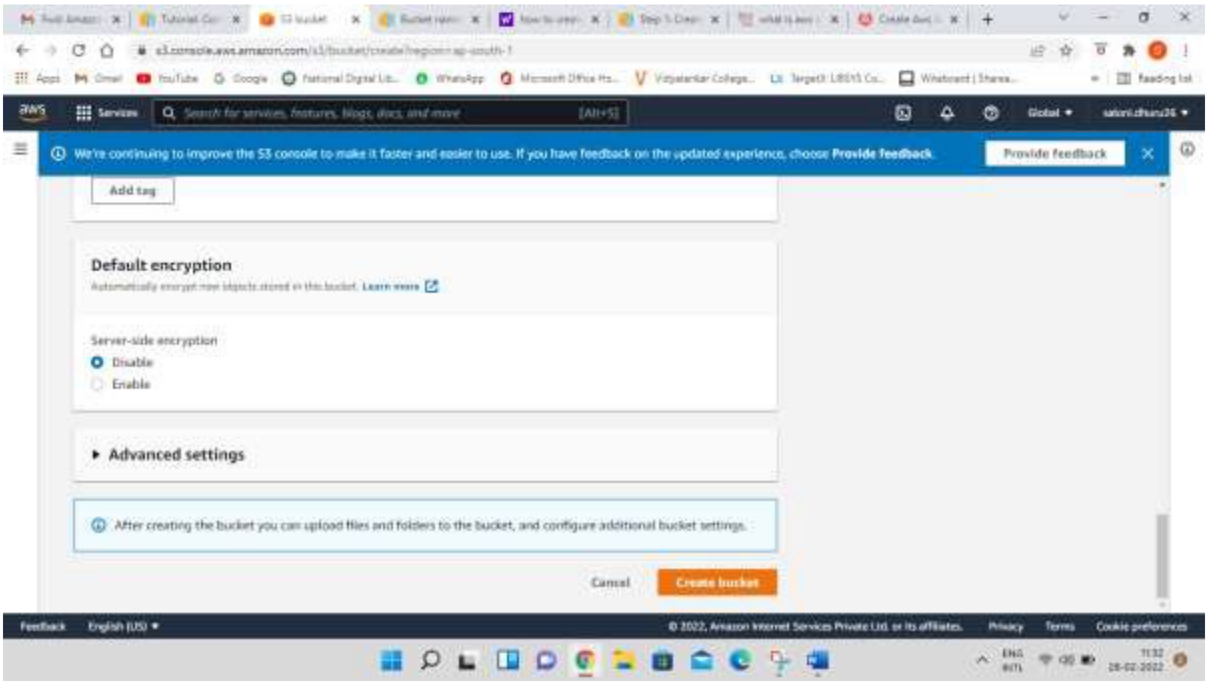
Step-5: Select the checkbox for **Turning off block all public access might result in this bucket and the objects within becoming public**



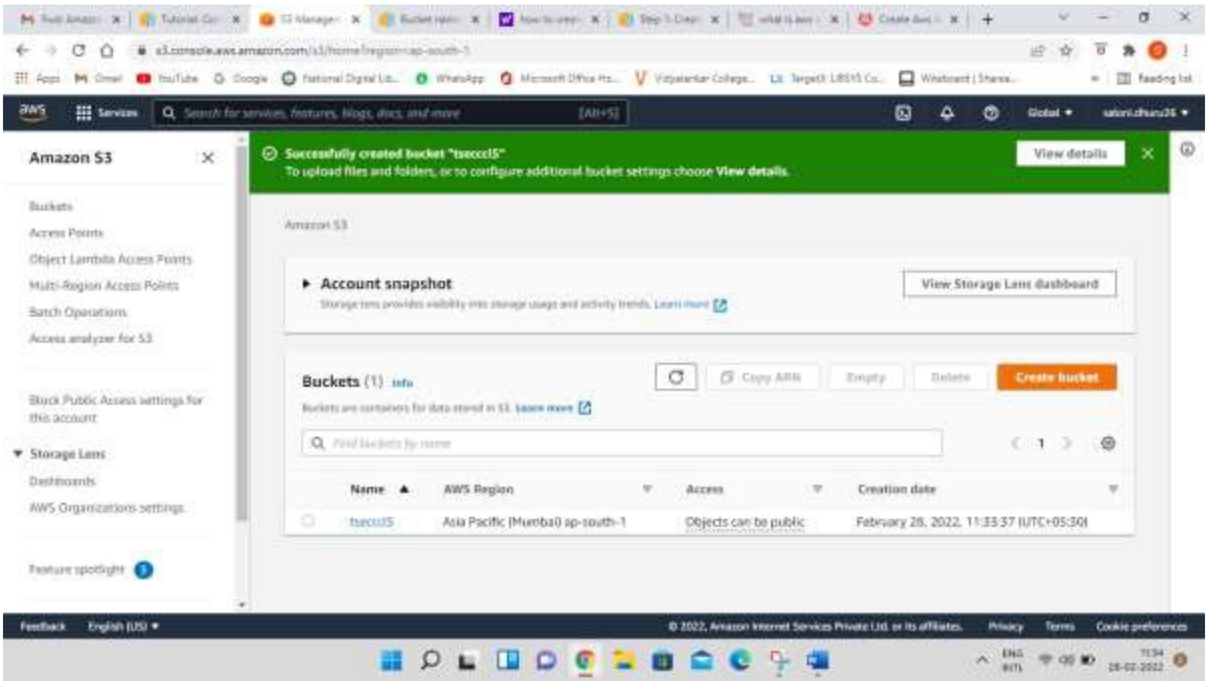
Step-6: Keep bucket versioning as disabled and add tags if required.



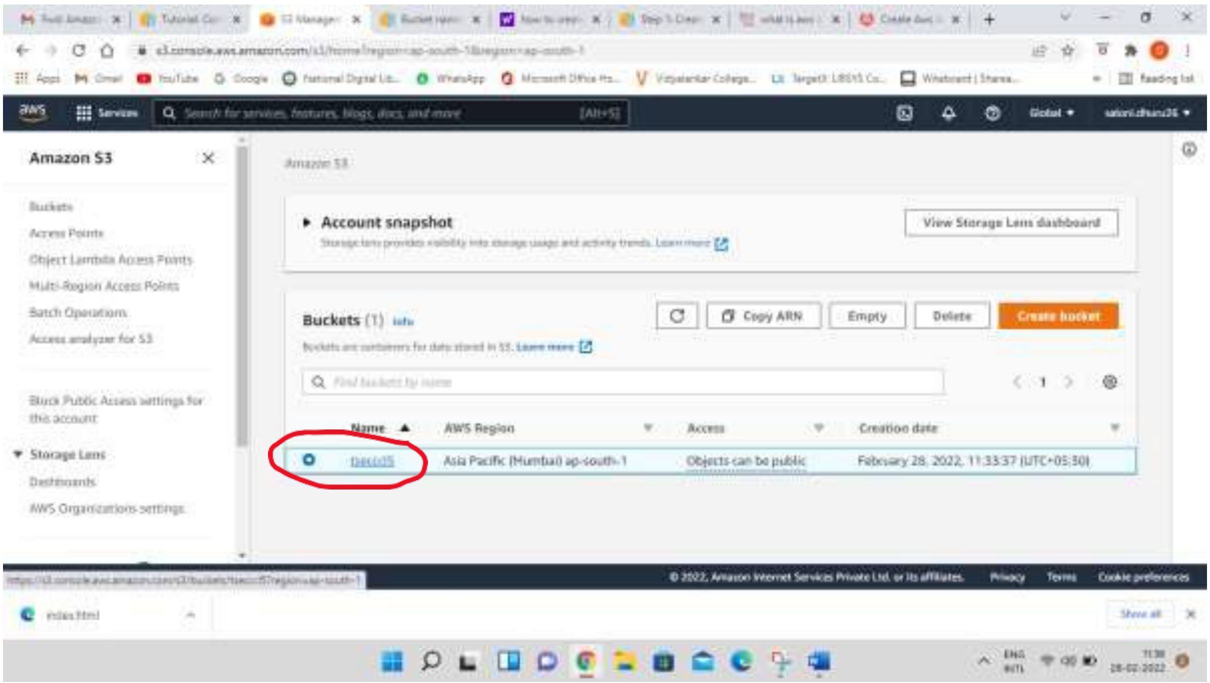
Step-7: Keep default encryption disabled and click on create bucket button



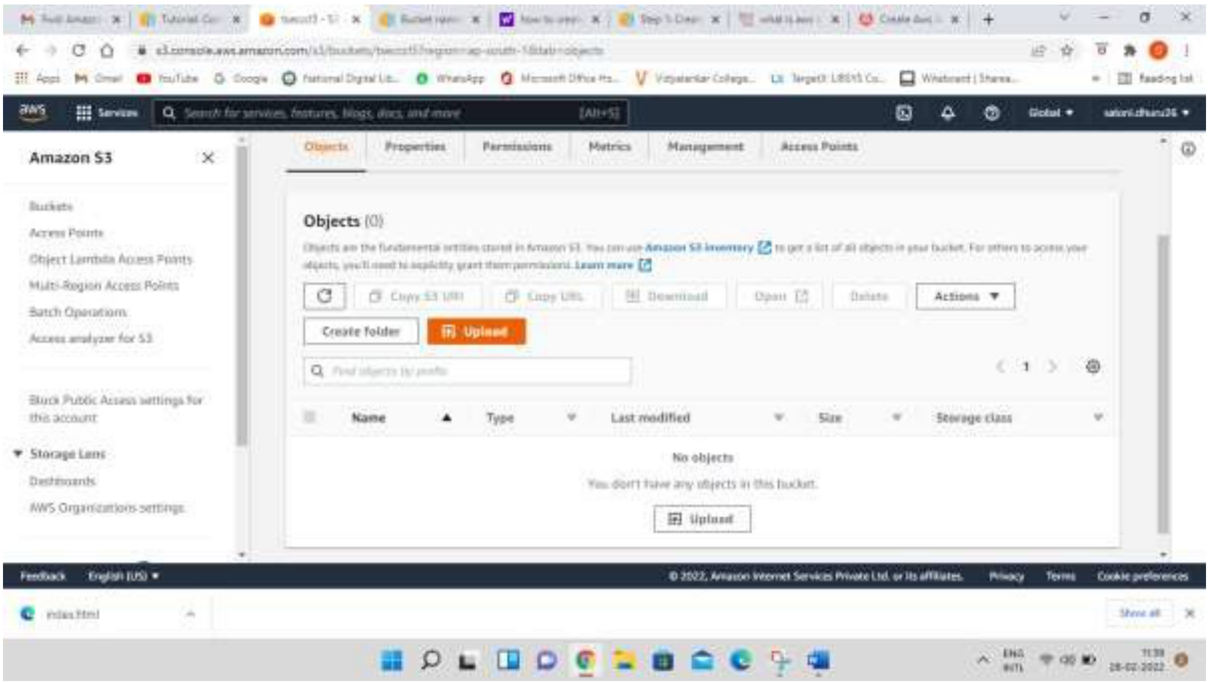
You can now see the successful creation of your bucket



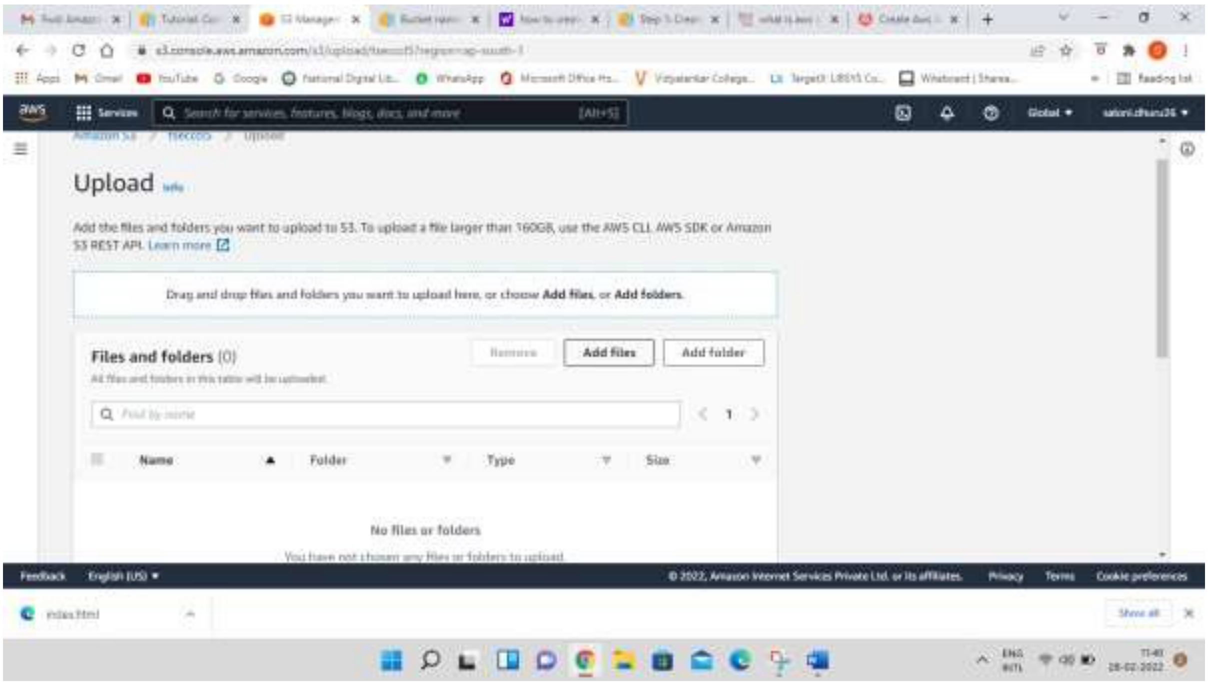
Step-8: now click on the bucket that you have created



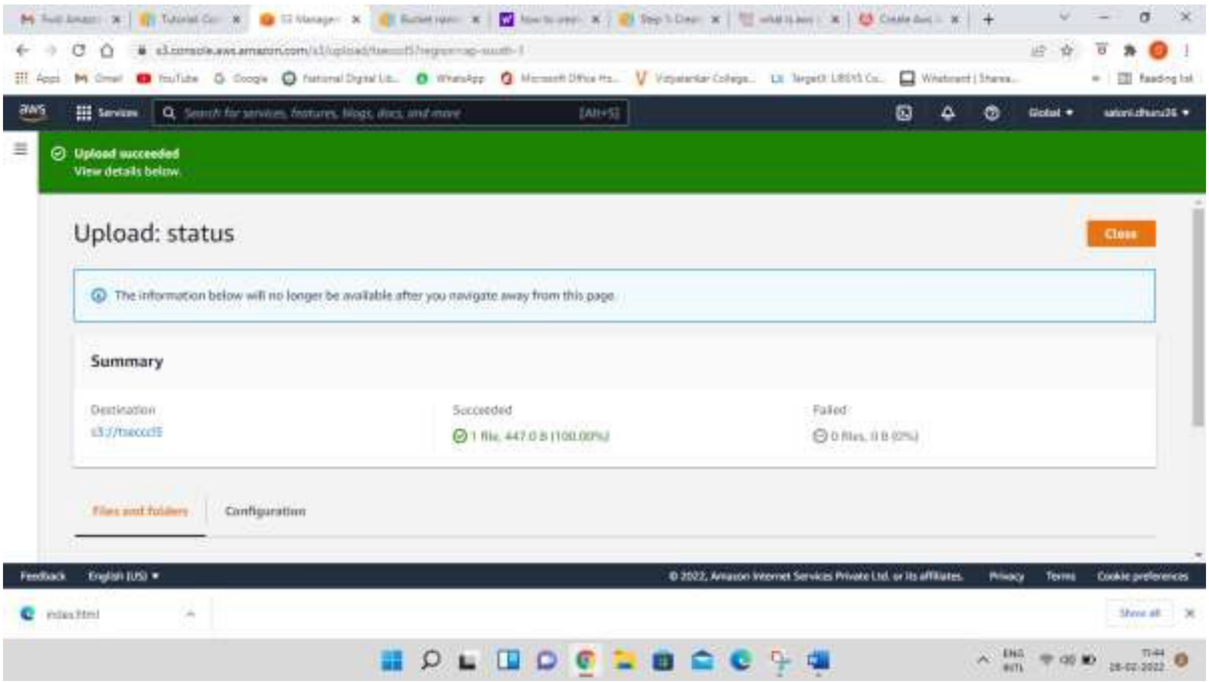
Step-9: You can either create a folder here or upload an existing file in the bucket



Step-10: now click on upload button and click on add files button browse your local machine and select which file you need to upload on S3 next click on upload button at bottom right end

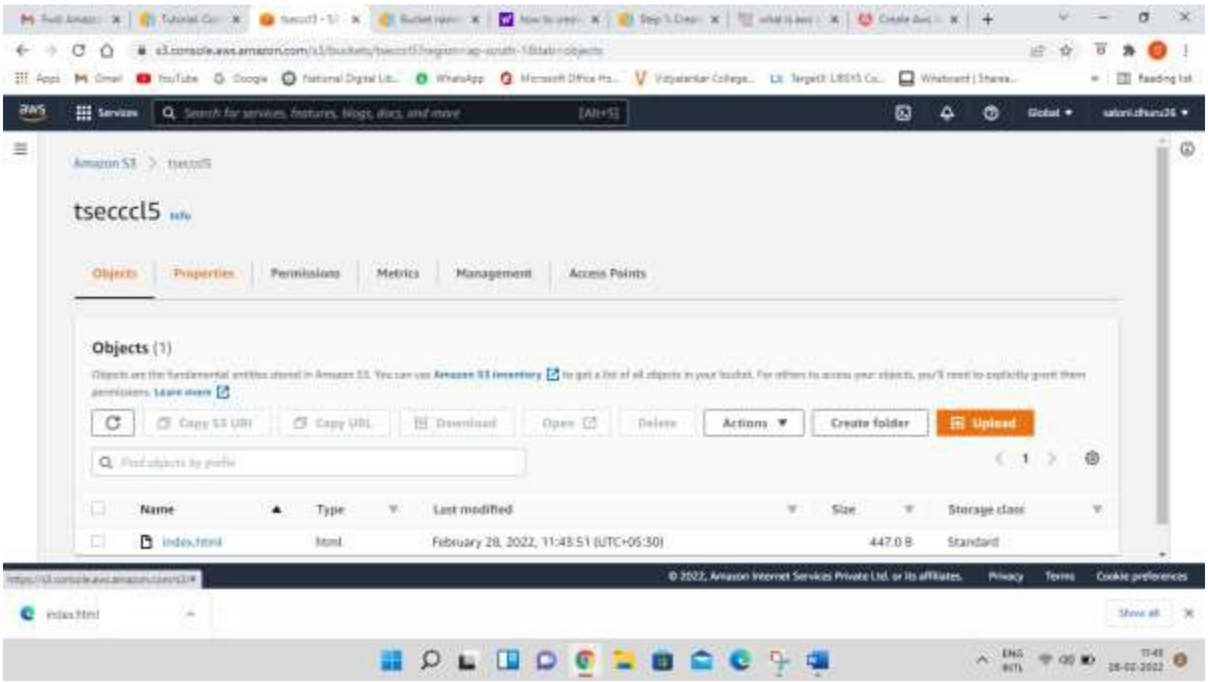


Now you can check the upload status screen

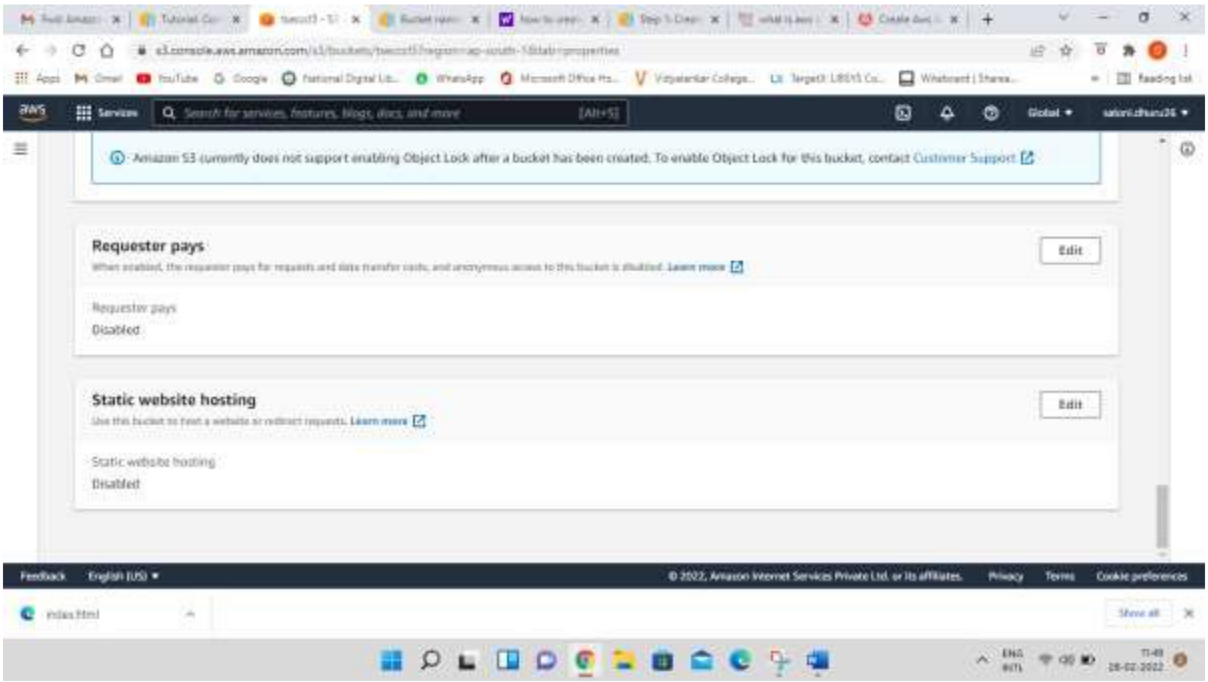


Now click on close button

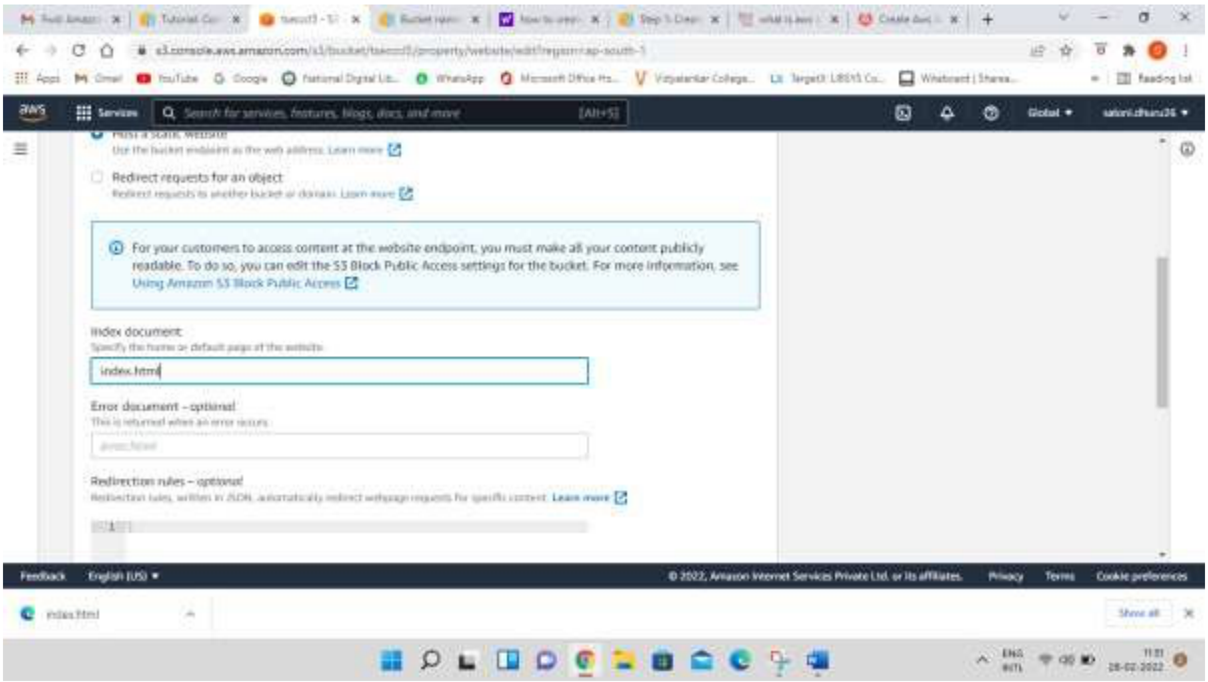
The screen will appear as below



Step-11: Select properties and scroll down to **Static website hosting** option which is disabled now click on Edit option on right side

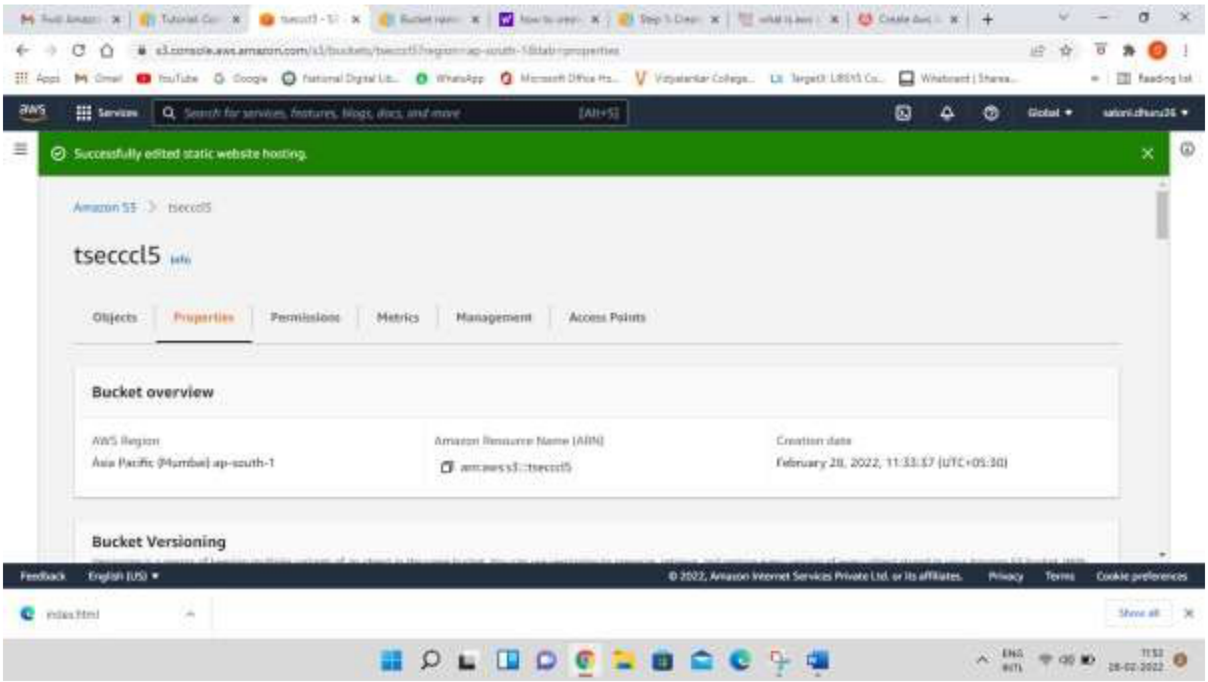


Step-12: Enable the radio button and specify the file name in **Index document** which you have added in S3

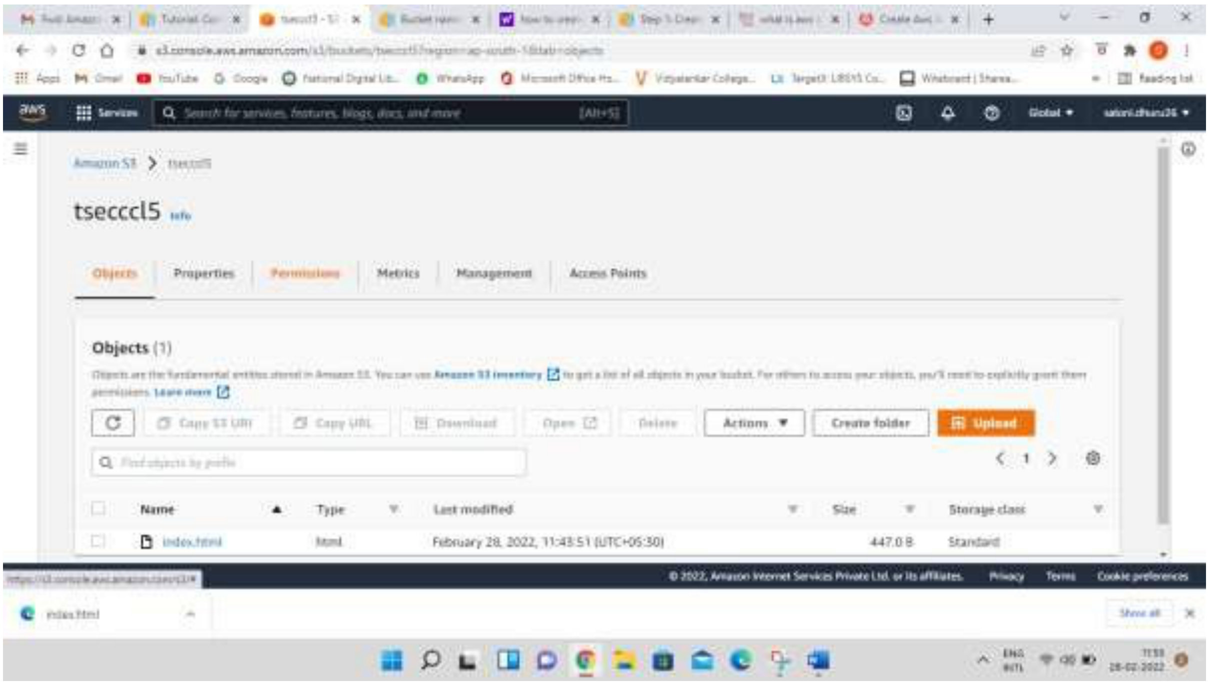


Scroll down and save the changes at bottom right

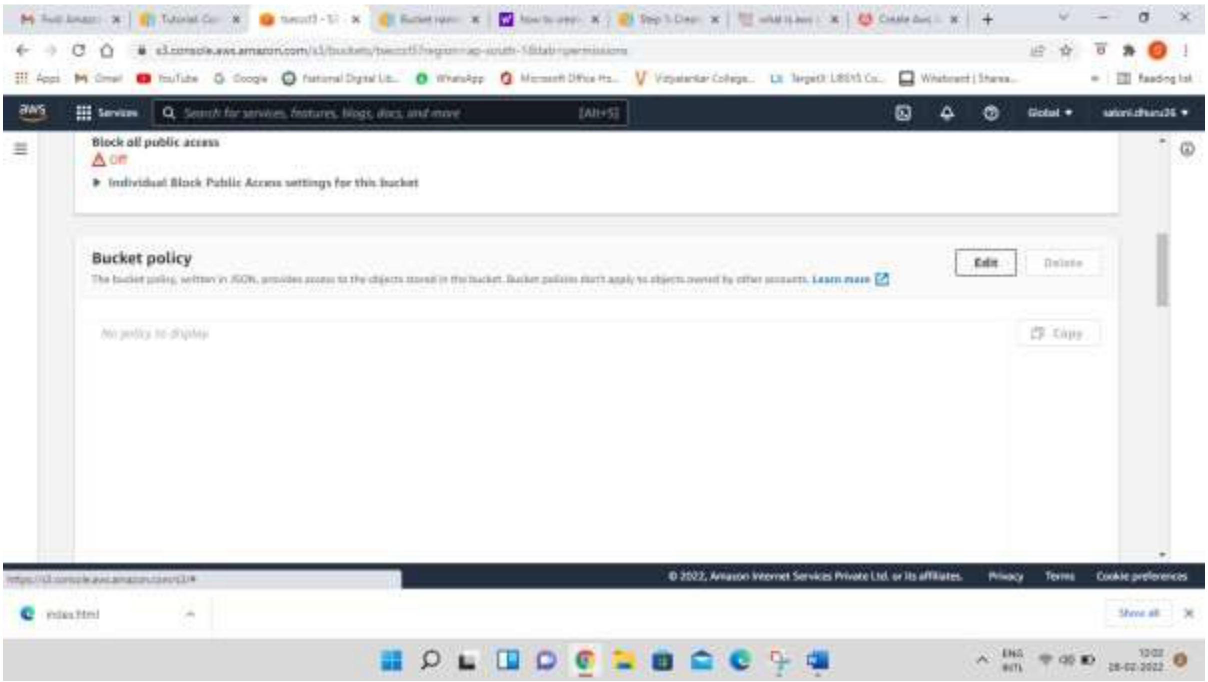
Following screen will appear



Step-13: Click on Permissions Tab



Step-14: In **bucket policy** click on Edit option

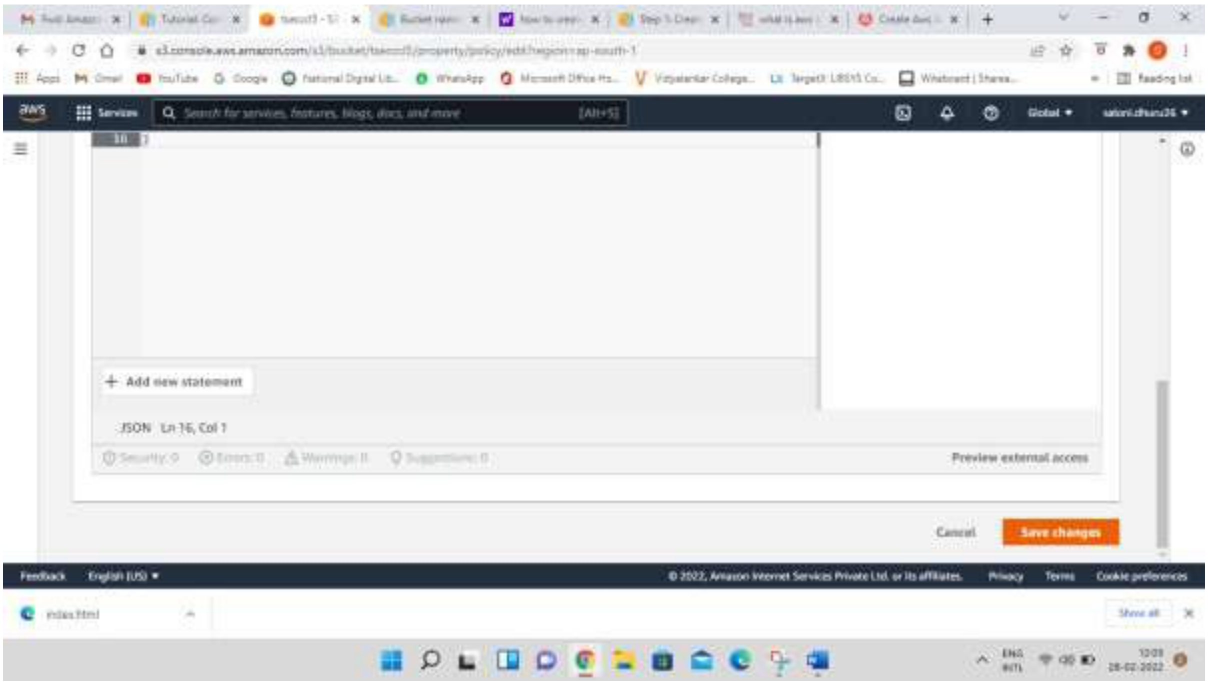


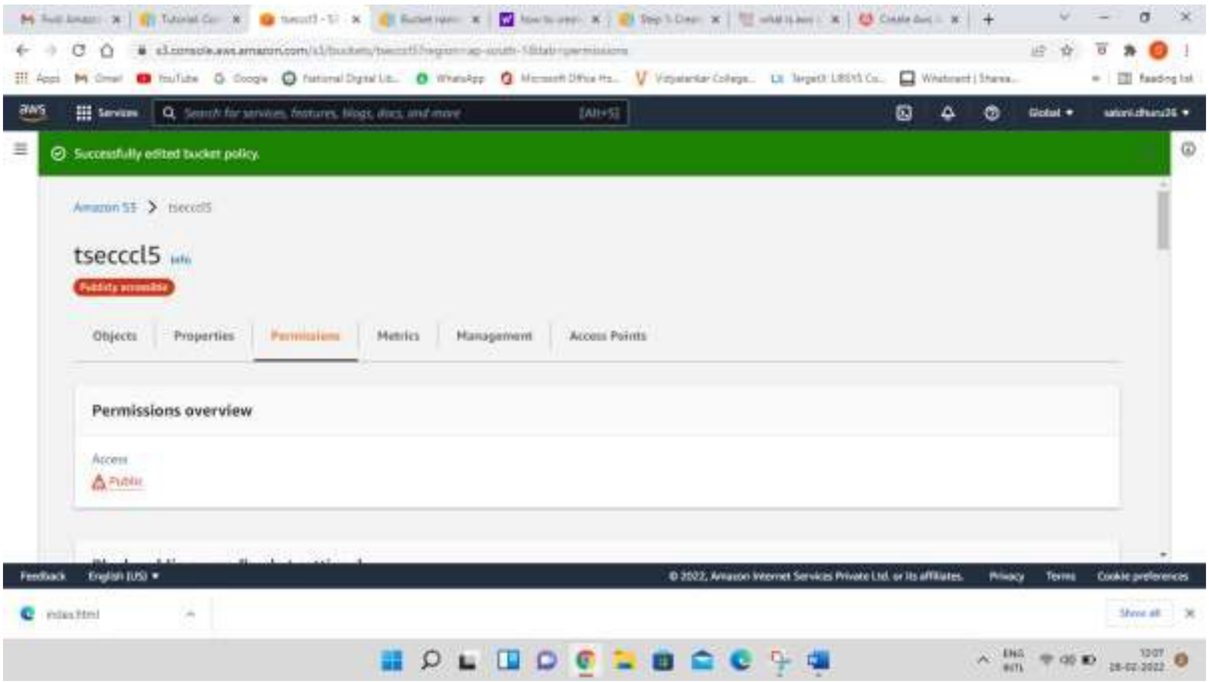
Step 15- after clicking on edit button paste the following code in bucket policy

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": [
        "s3:GetObject"
      ],
      "Resource": [
        "arn:aws:s3:::Bucket-Name/*"
      ]
    }
  ]
}
```

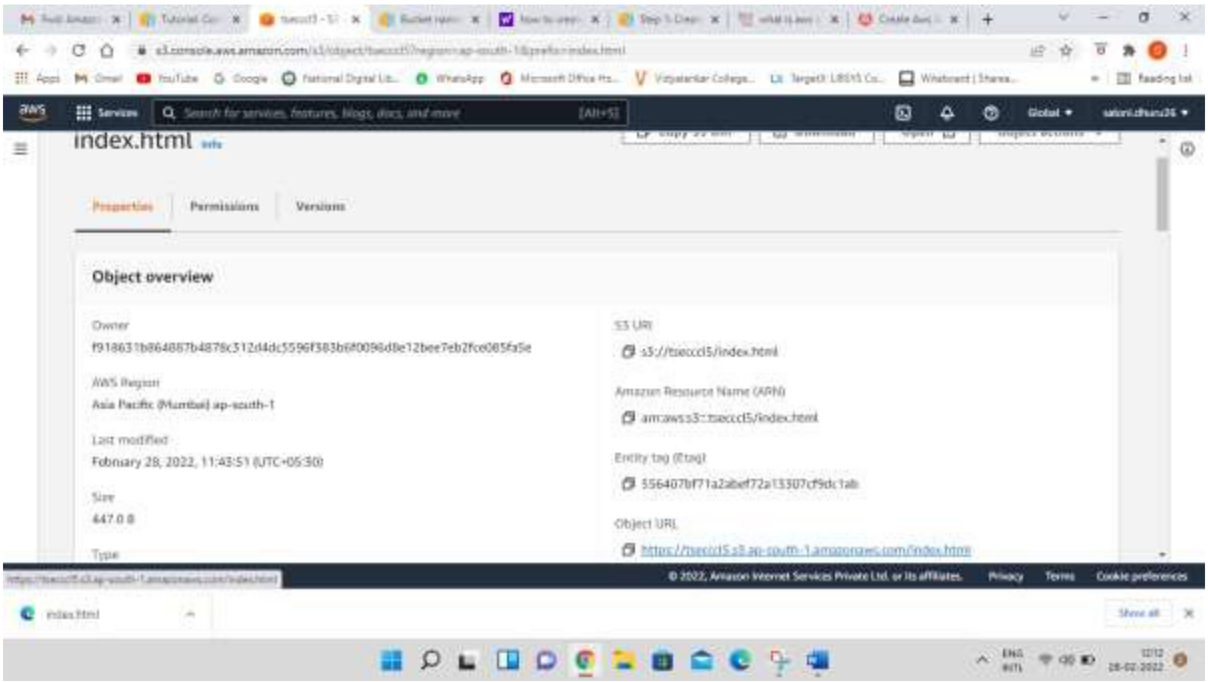
Note-Make sure that you add your bucket name in the code above

Scroll down and click on Save Changes button

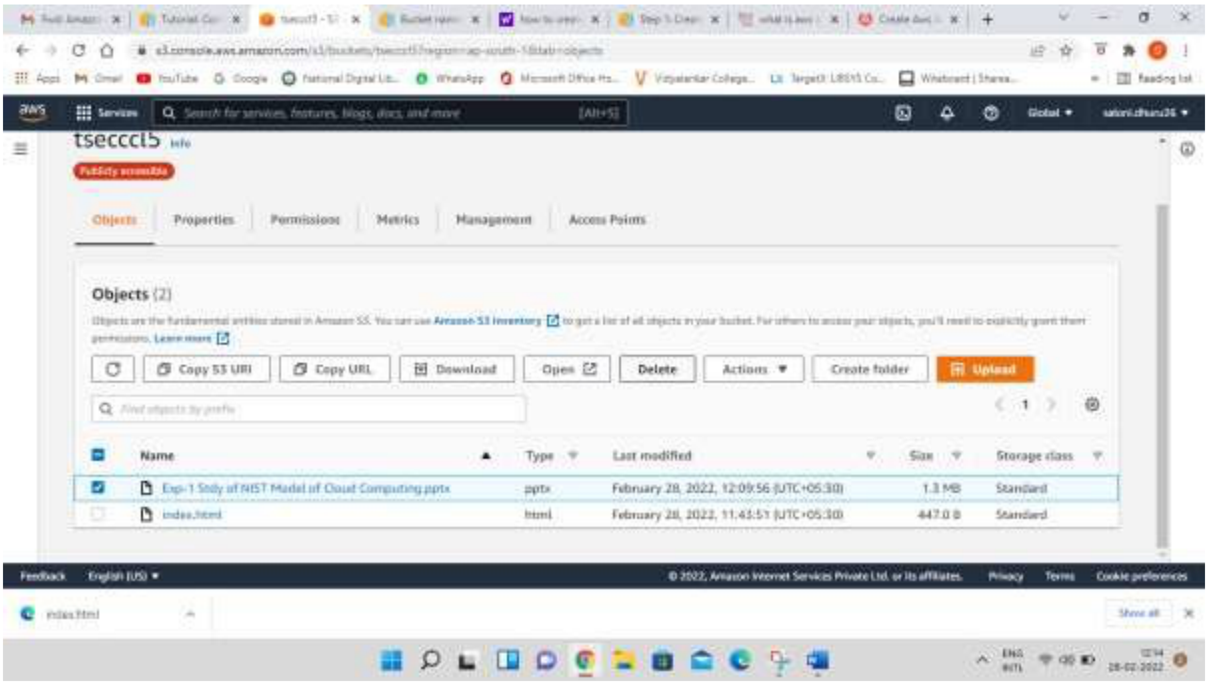




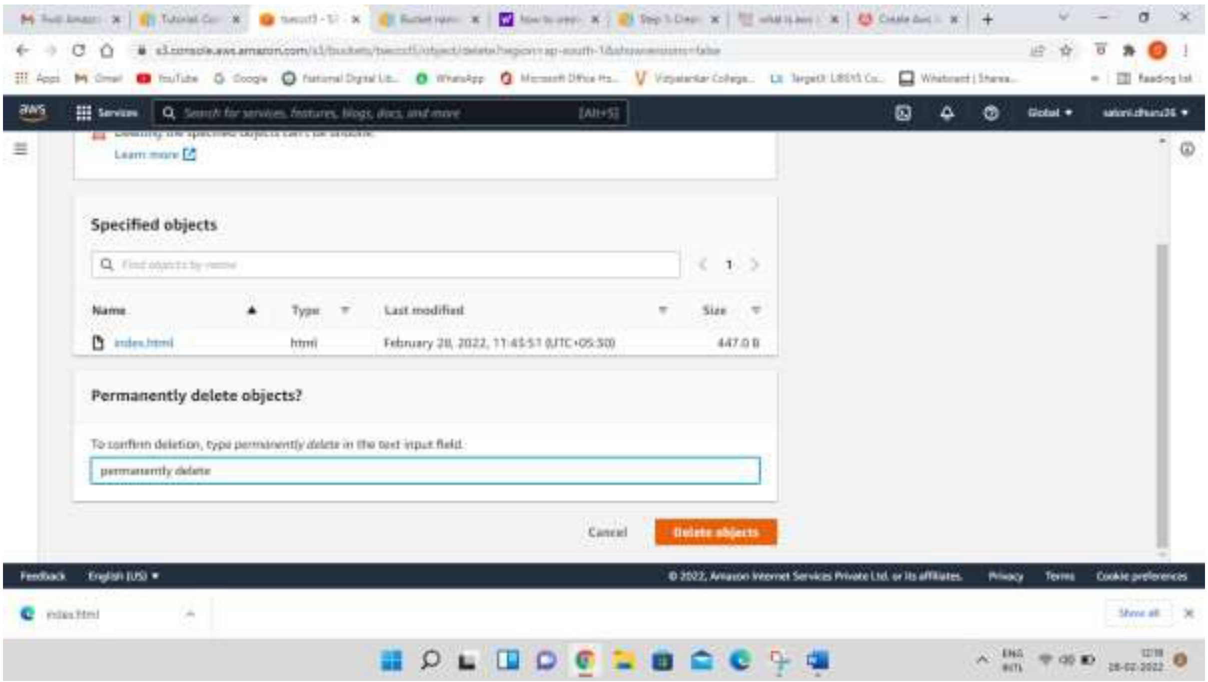
Step-16: open your html file and click on Object URL



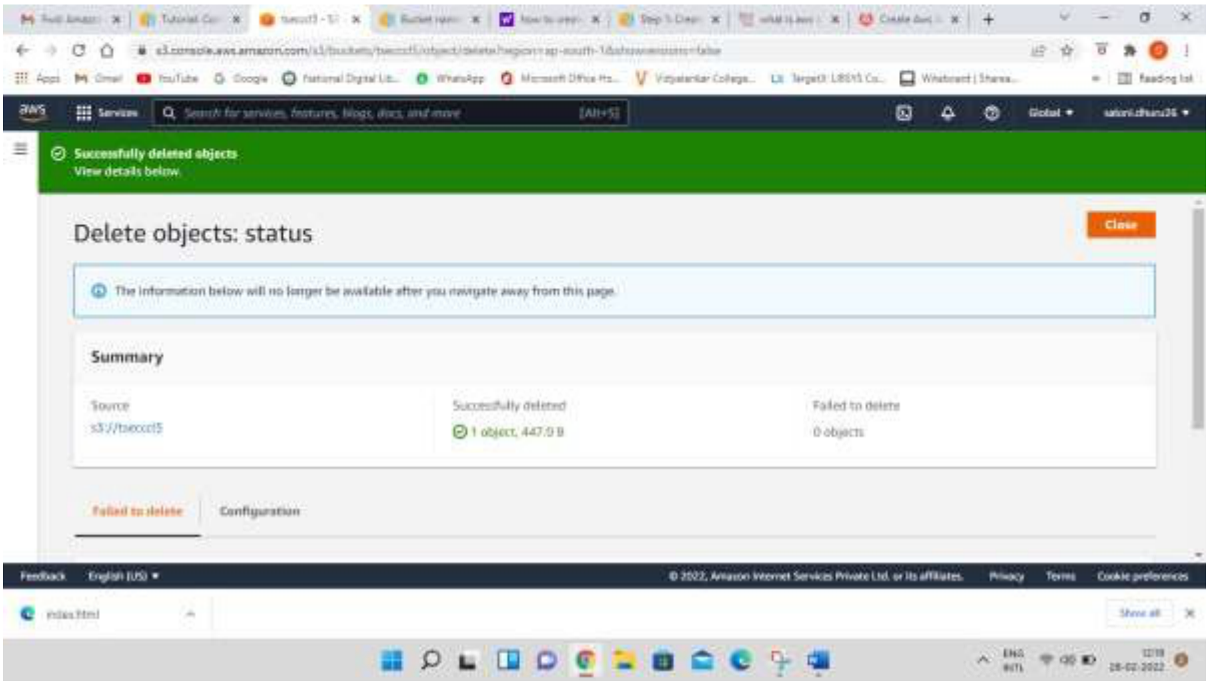
Step-17: Now for delete files click on checkbox of your file and then click on **Delete** Button



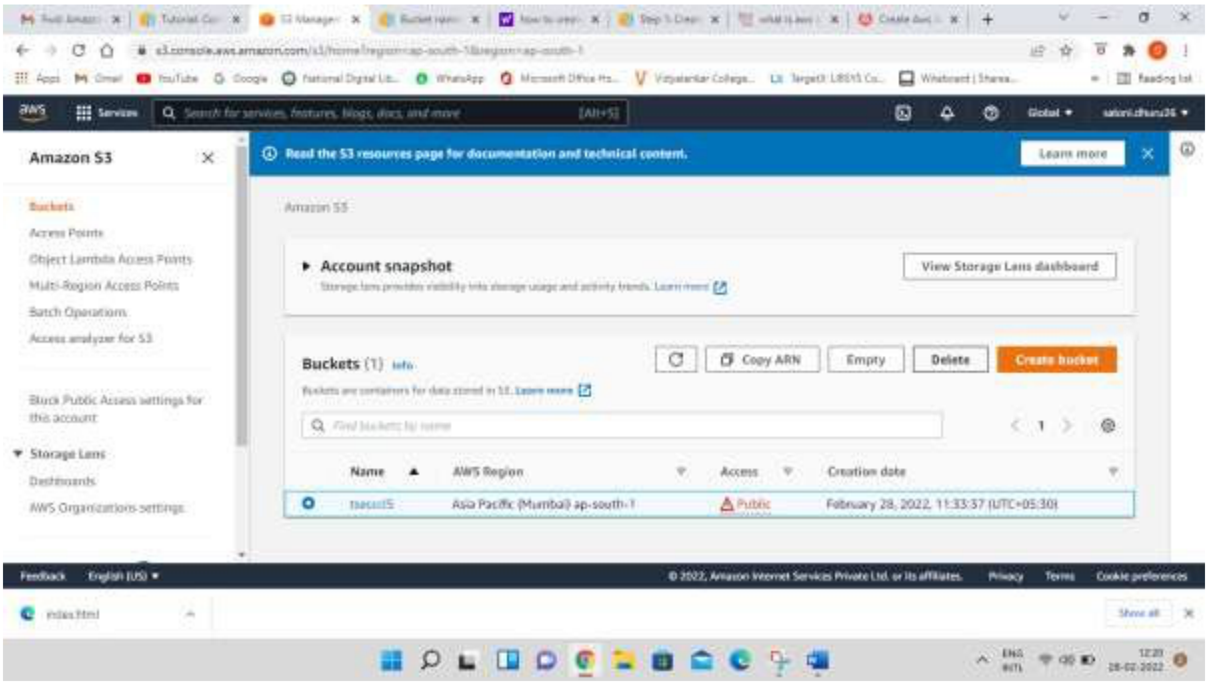
Write permanently delete and click on delete object button



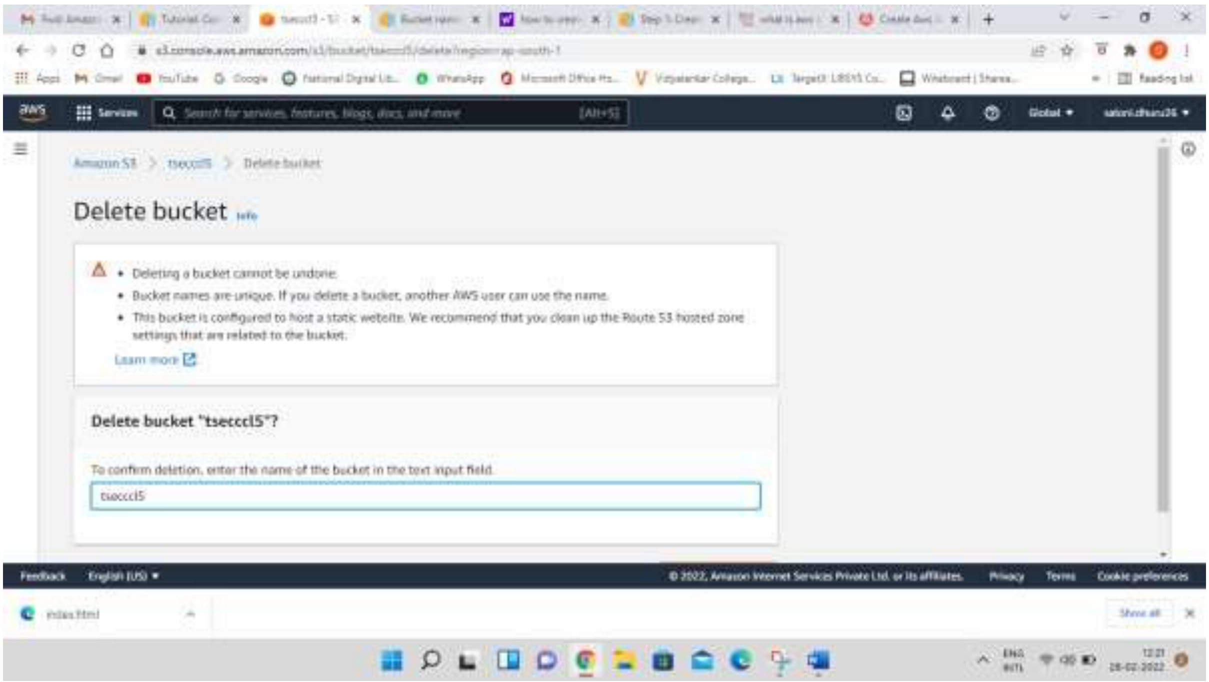
Now click on close button



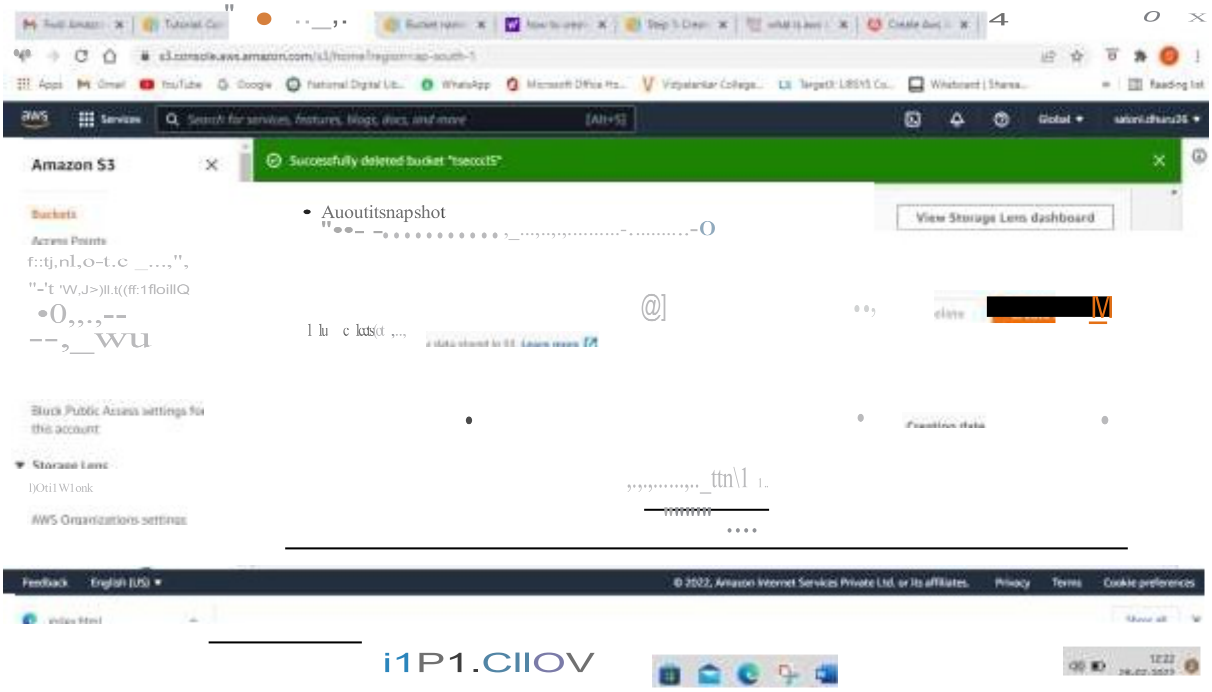
Step-18: now come to Amazon S3 tab and select your bucket and then click on delete button



Write down your bucket name in delete bucket tab and click on delete button at bottom right



You can see that the bucket is deleted


Services
Global
Amazon S3
Buckets
Access Points
Block Public Access settings for this account
Storage Lens
AWS Organizations settings
View Storage Lens dashboard
Creation date
Feedback
English (US)
© 2022, Amazon Internet Services Private Ltd. or its affiliates.
Privacy
Terms
Cookie preferences

PART B

(PART B: TO BE COMPLETED BY STUDENTS)

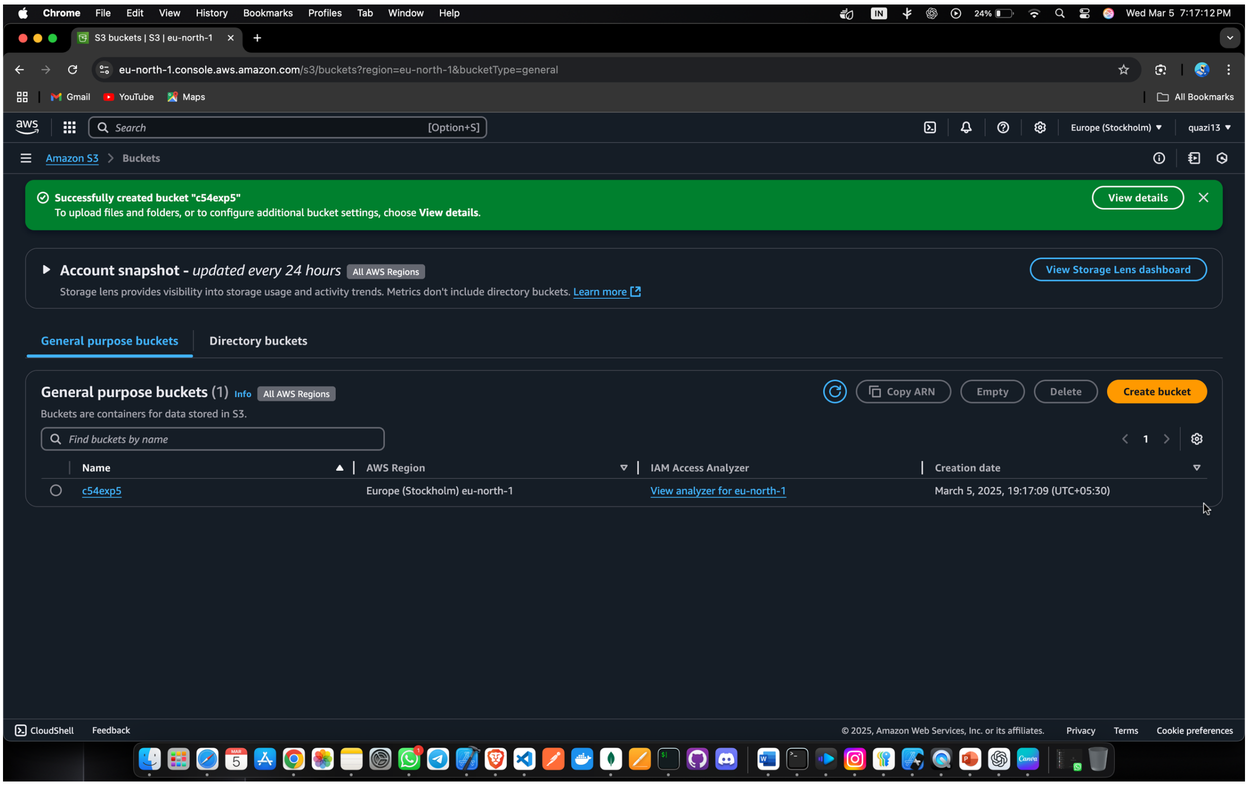
(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the ERP or emailed to the concerned lab in charge faculties at the end of the practical in case the there is no ERP access available)

Roll No.C54	Name: Quazi Md Moinuddin
Class :TE-C	Batch :C3
Date of Experiment:	Date of Submission:5-2-25
Grade :	

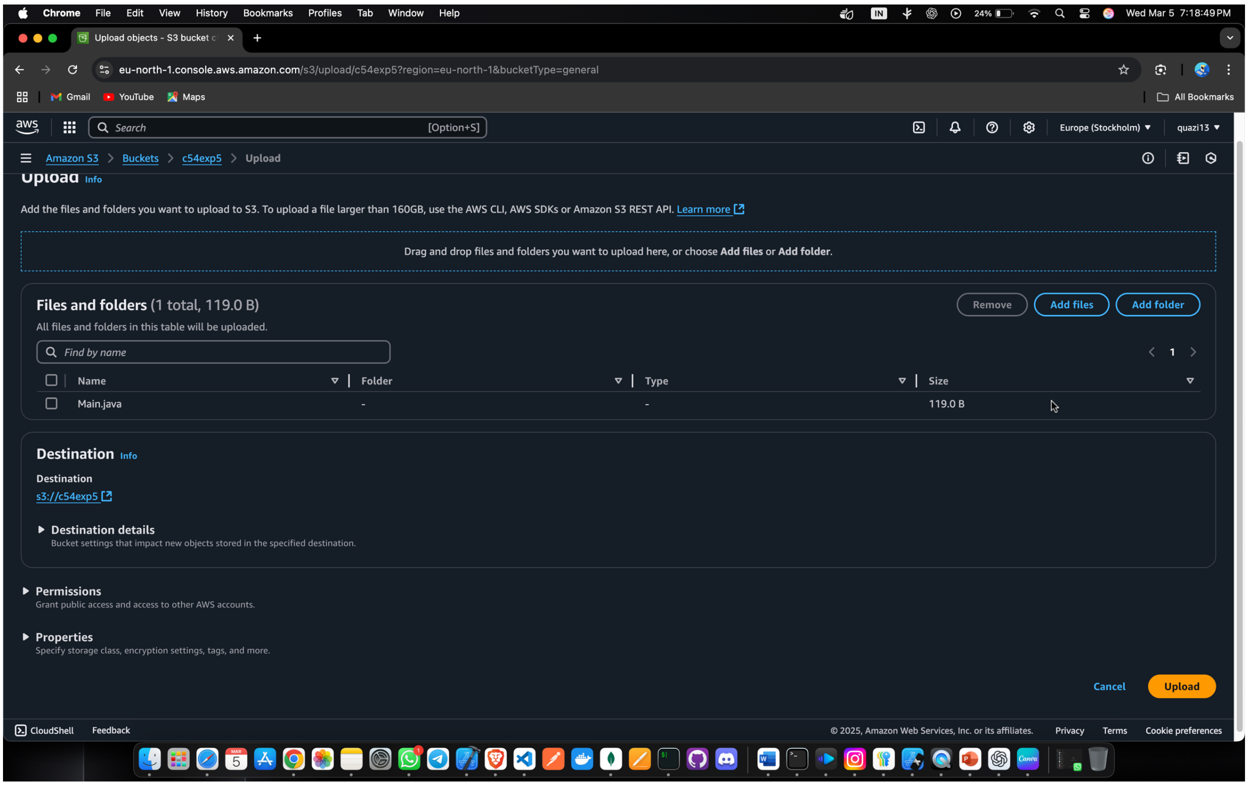
B.1 Question of Curiosity:

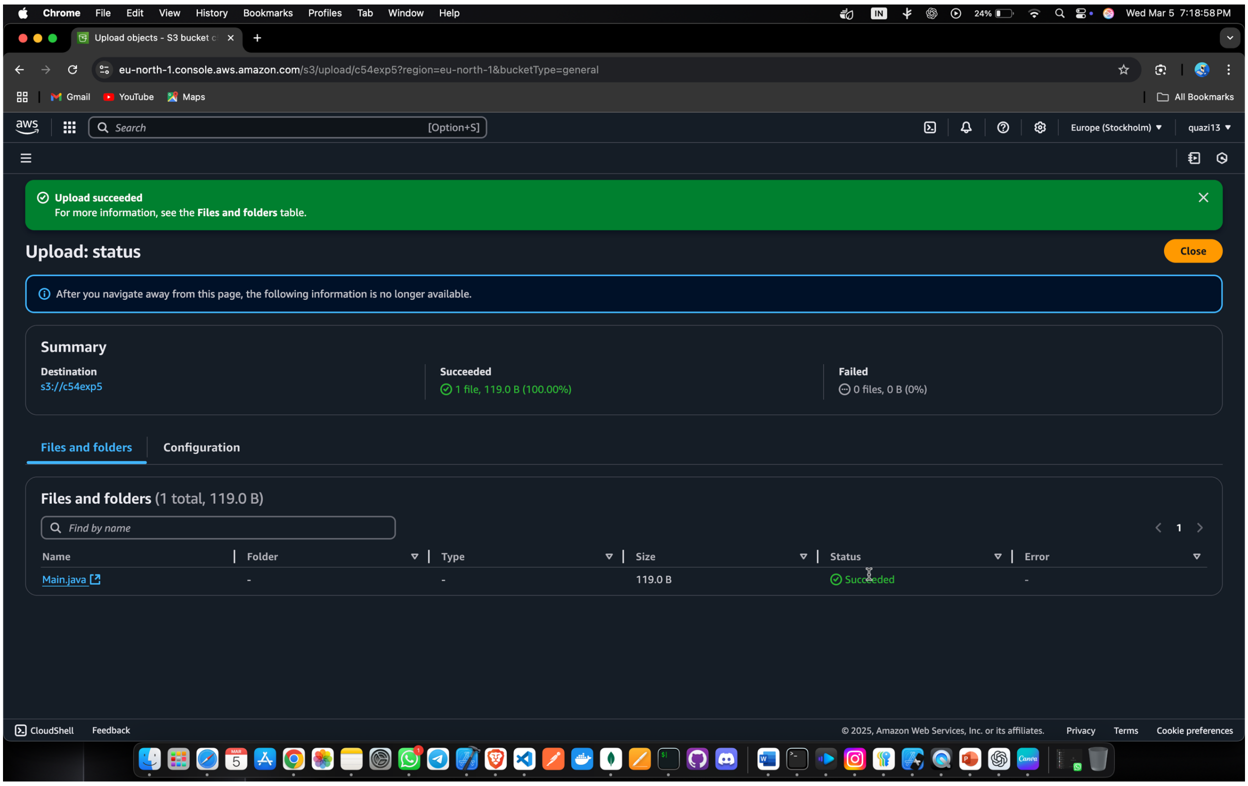
(To be answered by student based on the practical performed and learning/observations)

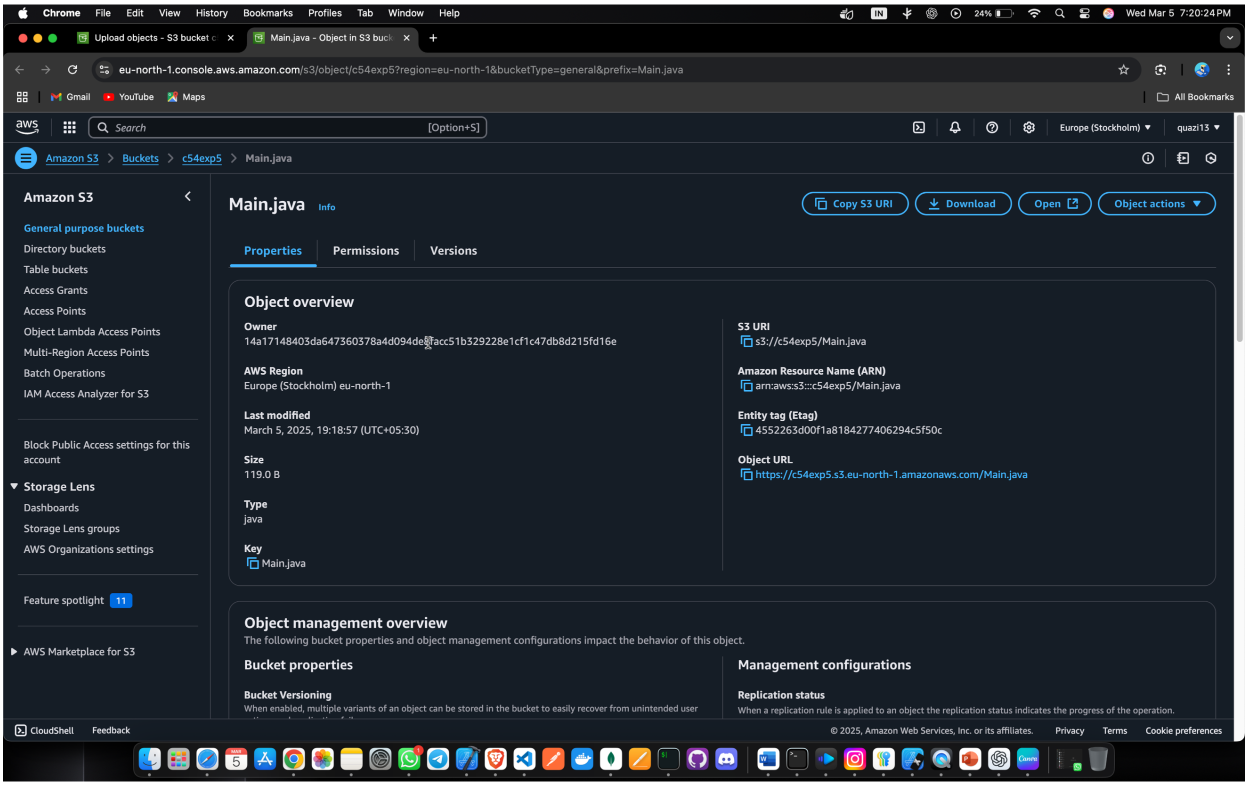
Q.1: Create Bucket using AWS S3 service (Add stepwise screenshots of the same)



Q2: Add Objects to *Bucket created (Add stepwise screenshots of the same)*


Upload objects - S3 bucket c
eu-north-1.console.aws.amazon.com/s3/upload/c54exp5?region=eu-north-1&bucketType=general
Amazon S3 > Buckets > c54exp5 > Upload
Upload Info
Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. Learn more
Drag and drop files and folders you want to upload here, or choose Add files or Add folder.
Files and folders (1 total, 119.0 B)
All files and folders in this table will be uploaded.
Remove
Add files
Add folder
Find by name
Name
Folder
Type
Size
Main.java
119.0 B
Destination Info
Destination
s3://c54exp5
Destination details
Bucket settings that impact new objects stored in the specified destination.
Permissions
Grant public access and access to other AWS accounts.
Properties
Specify storage class, encryption settings, tags, and more.
Cancel
Upload
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Upload objects - S3 bucket c
eu-north-1.console.aws.amazon.com/s3/upload/c54exp5?region=eu-north-1&bucketType=general
Upload succeeded
For more information, see the Files and folders table.
Upload: status
Close
After you navigate away from this page, the following information is no longer available.
Summary
Destination
s3://c54exp5
Succeeded
1 file, 119.0 B (100.00%)
Failed
0 files, 0 B (0%)
Files and folders
Configuration
Files and folders (1 total, 119.0 B)
Find by name
Name
Folder
Type
Size
Status
Error
Main.java
119.0 B
Succeeded
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Chrome File Edit View History Bookmarks Profiles Tab Window Help
Wed Mar 5 7:20:24 PM
Upload objects - S3 bucket c
Main.java - Object in S3 buck
eu-north-1.console.aws.amazon.com/s3/object/c54exp5?region=eu-north-1&bucketType=general&prefix=Main.java
Gmail YouTube Maps
All Bookmarks
Search [Option+S]
Europe (Stockholm)
quazi13
Amazon S3 > Buckets > c54exp5 > Main.java
Amazon S3
General purpose buckets
Directory buckets
Table buckets
Access Grants
Access Points
Object Lambda Access Points
Multi-Region Access Points
Batch Operations
IAM Access Analyzer for S3
Block Public Access settings for this account
Storage Lens
Dashboards
Storage Lens groups
AWS Organizations settings
Feature spotlight 11
AWS Marketplace for S3
Main.java Info
Copy S3 URI
Download
Open
Object actions
Properties
Permissions
Versions
Object overview
Owner
14a17148403da647360378a4d094de8facc51b329228e1cf1c47db8d215fd16e
AWS Region
Europe (Stockholm) eu-north-1
Last modified
March 5, 2025, 19:18:57 (UTC+05:30)
Size
119.0 B
Type
java
Key
Main.java
S3 URI
s3://c54exp5/Main.java
Amazon Resource Name (ARN)
arn:aws:s3:::c54exp5/Main.java
Entity tag (Etag)
4552263d00f1a8184277406294c5f50c
Object URL
https://c54exp5.s3.eu-north-1.amazonaws.com/Main.java
Object management overview
The following bucket properties and object management configurations impact the behavior of this object.
Bucket properties
Bucket Versioning
When enabled, multiple variants of an object can be stored in the bucket to easily recover from unintended user
Management configurations
Replication status
When a replication rule is applied to an object the replication status indicates the progress of the operation.
CloudShell
Feedback
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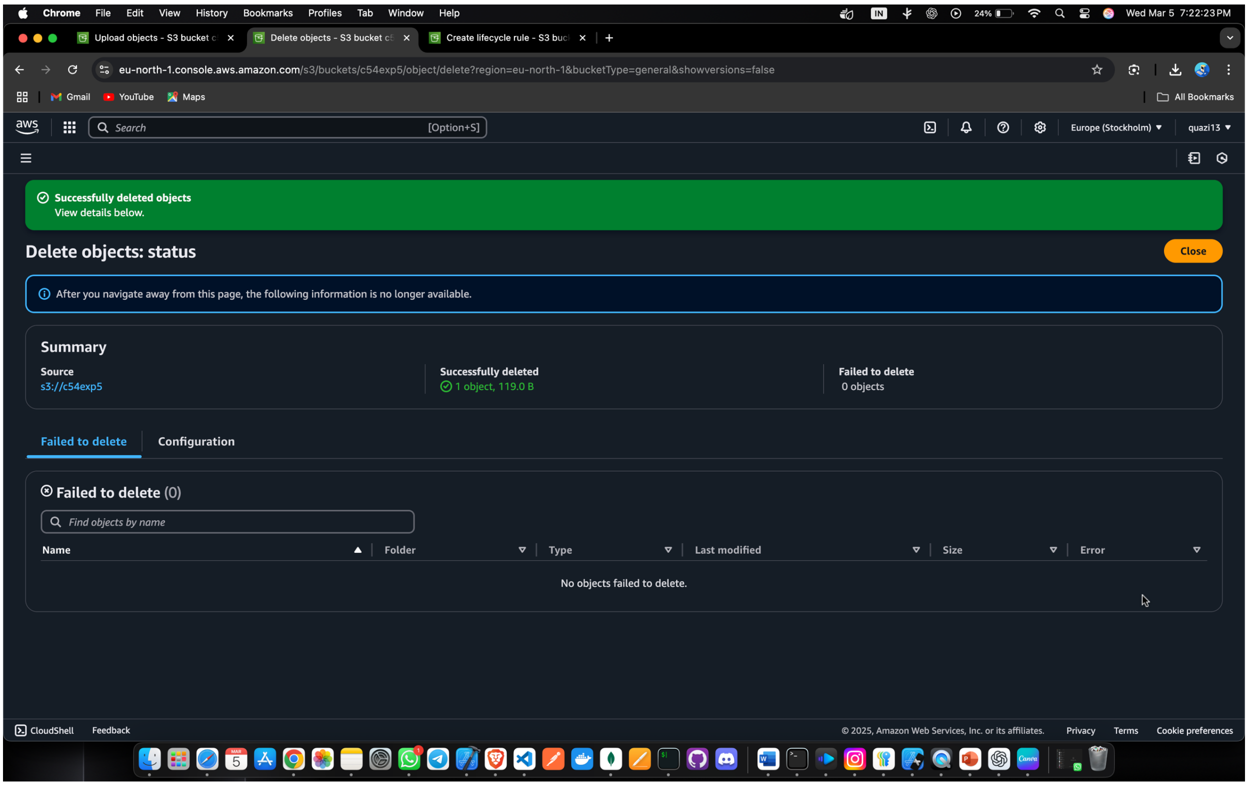
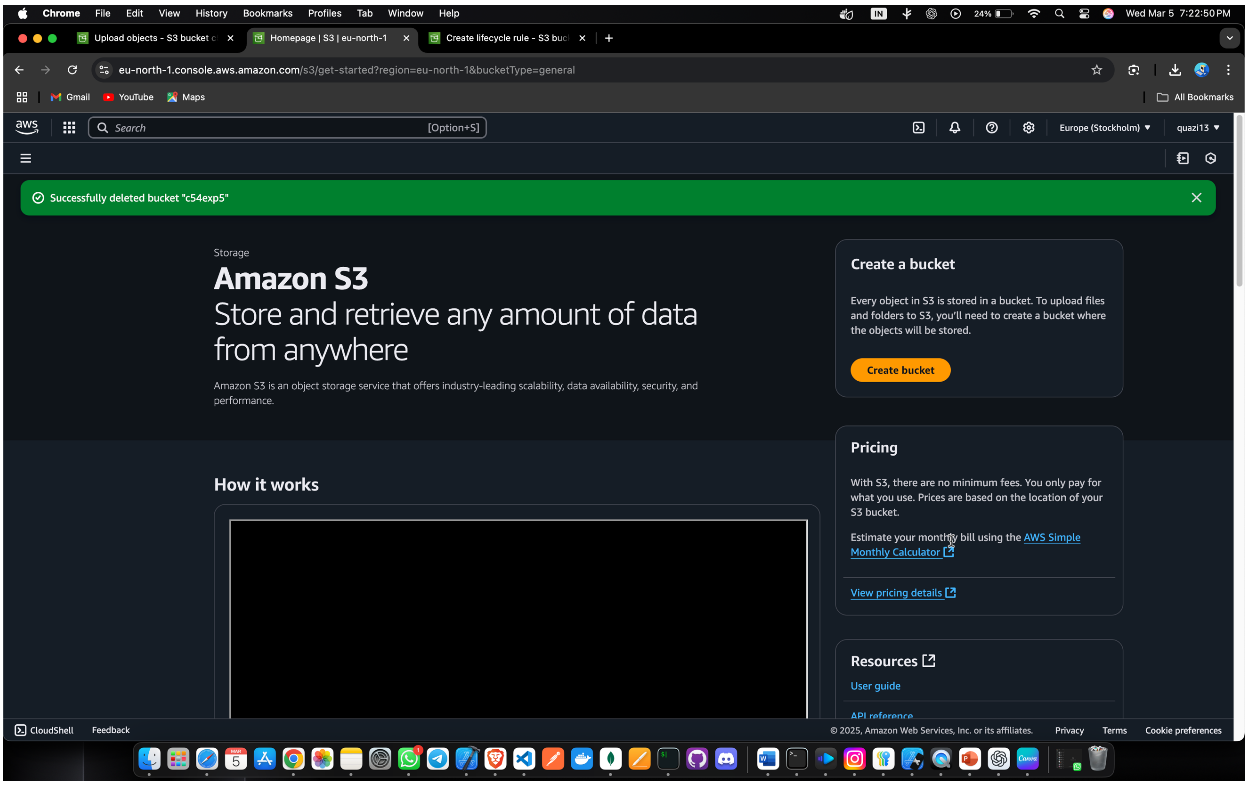

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Wed Mar 5 7:22:23 PM
Upload objects - S3 bucket c
Delete objects - S3 bucket c
Create lifecycle rule - S3 buc
eu-north-1.console.aws.amazon.com/s3/buckets/c54exp5/object/delete?region=eu-north-1&bucketType=general&showversions=false
Gmail YouTube Maps
All Bookmarks
Search [Option+S]
Europe (Stockholm)
quazi13
Successfully deleted objects
View details below.
Delete objects: status
Close
After you navigate away from this page, the following information is no longer available.
Summary
Source
s3://c54exp5
Successfully deleted
1 object, 119.0 B
Failed to delete
0 objects
Failed to delete
Configuration
Failed to delete (0)
Find objects by name
Name
Folder
Type
Last modified
Size
Error
No objects failed to delete.
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences



Q3: Compare Google drive with AWS S3

AWS S3	AWS EBS
The AWS S3 Full form is Amazon Simple Storage Service	The AWS EBS full form is Amazon Elastic Block Store
AWS S3 is an object storage service that helps the industry in scalability, data availability, security, etc.	It is easy to use.
AWS S3 is used to store and protect any amount of data for a range of use cases.	It has high-performance block storage at every scale.
AWS S3 can be used to store data lakes, websites, mobile applications, backup and restore big data analytics. , enterprise applications, IoT devices, archive etc.	It is scalable.
AWS S3 also provides management features.	It is also used to run relational or NoSQL databases.

B.2 Conclusion:

(Students must write the conclusion as per the attainment of individual outcome

listed above and learning/observation)

In conclusion, experiment provided a comprehensive understanding of storage services in AWS, particularly focusing on Amazon S3. Through practical implementation, students gained insights into the features and use cases of AWS storage solutions, facilitating their ability to deploy Infrastructure as a Service (IAAS) models in cloud computing. This experiment underscored the importance of storage management in cloud environments and equipped students with essential skills for leveraging AWS storage services effectively.